

# CSE 4417 IoT Theory

## Lecture 08: IoT Middleware

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Class Code  
bjedchaj

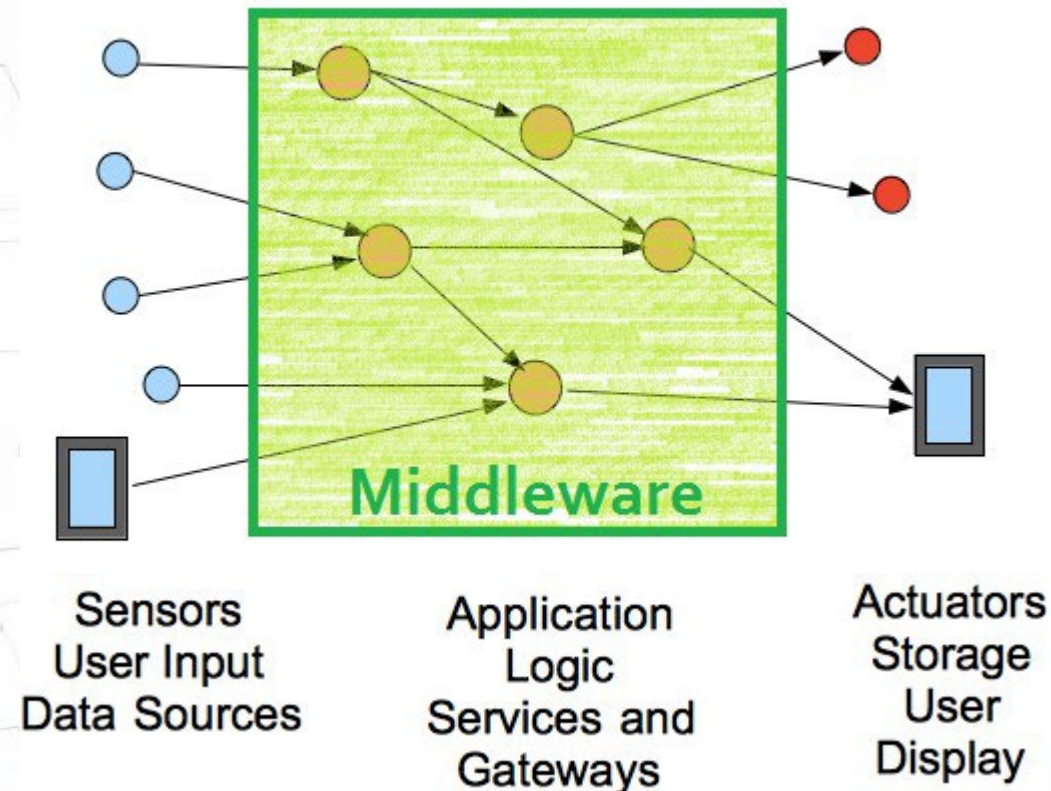


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# Middleware – or, haven't we covered this already?

## IoT Applications as a Graph



- Communication
- Storage
- Data management
- Software
- Platforms

# So why differentiate middleware?

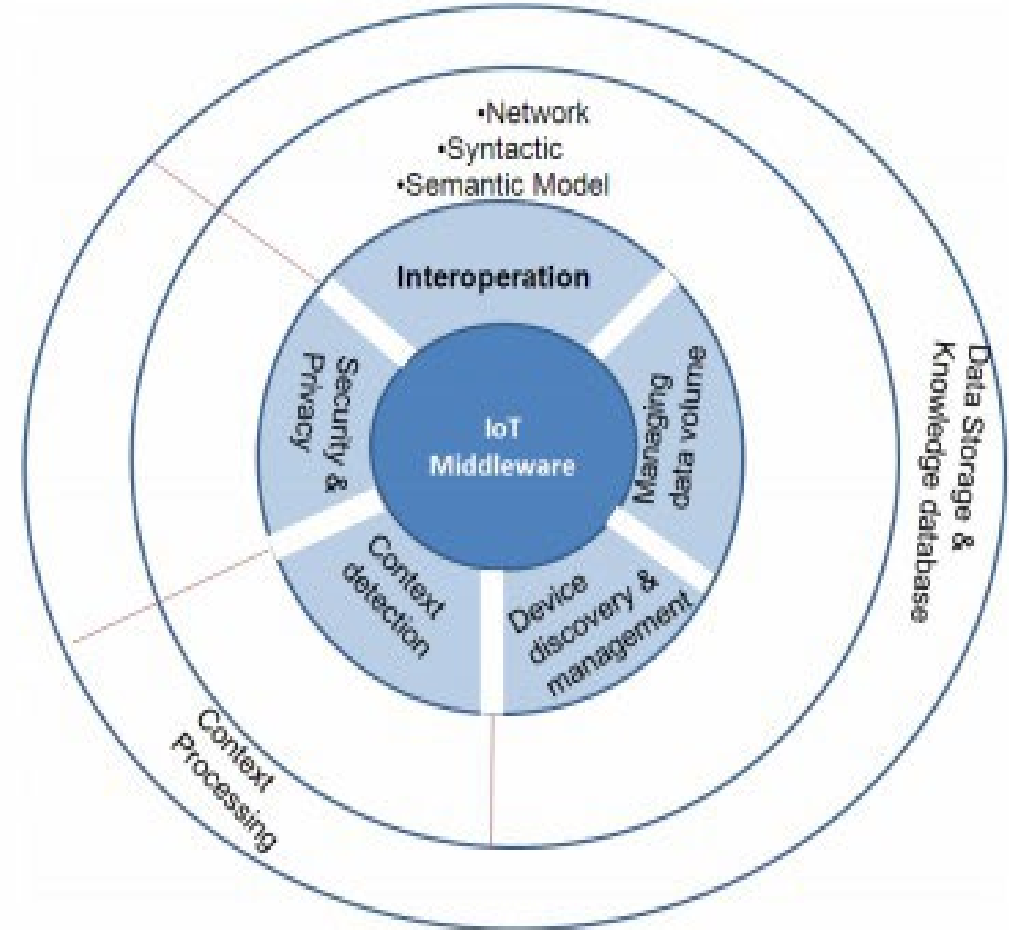
- Unify/standardize protocols
- Infrastructure “glue” to provide abstraction and layers
- API/Interfaces:
  - Connect with existing middleware components
  - Establish interfaces for a component to be developed/modified





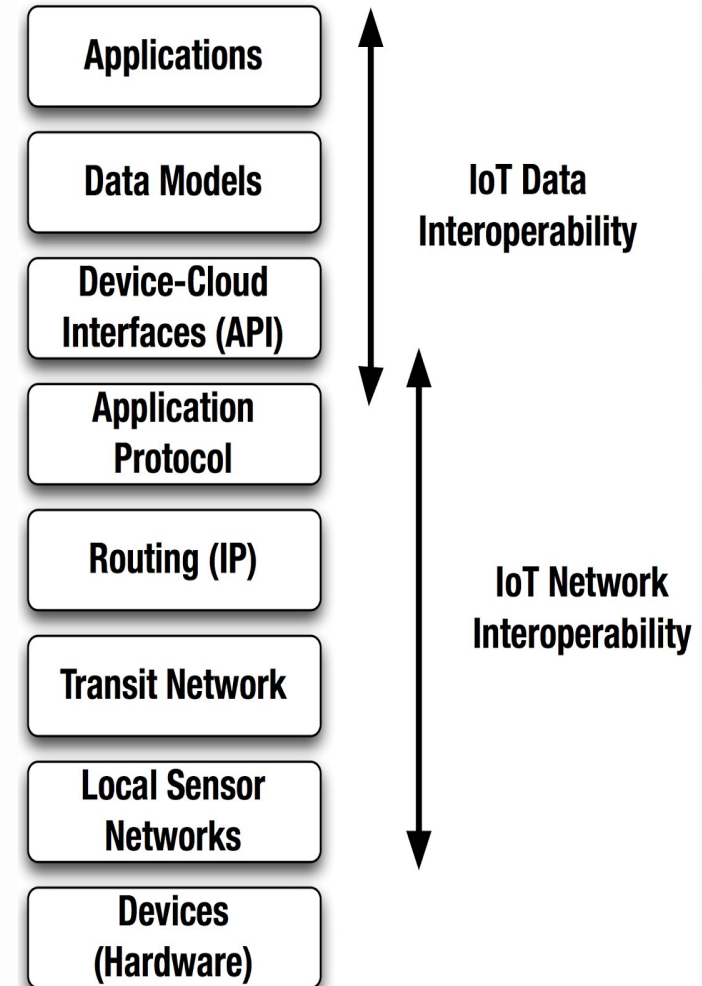
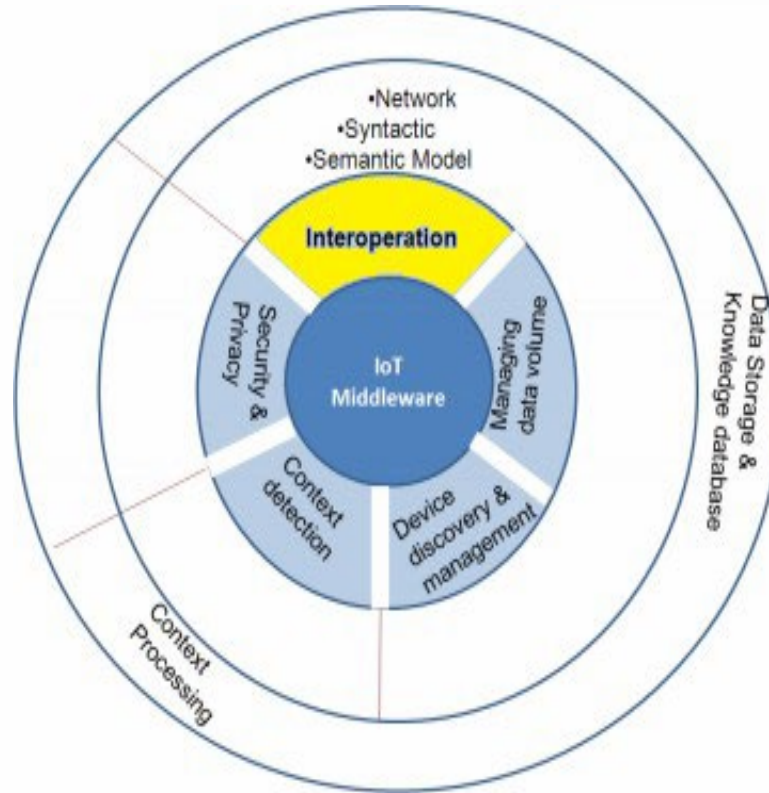
# Functional middleware building blocks

- Interoperability
- Security/Privacy
- Context Detection
- Device Discovery
- Scalability



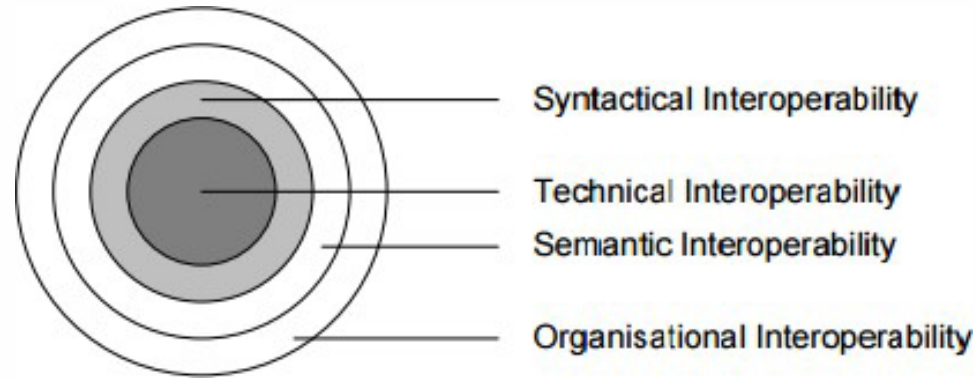
# Interoperation

- Network
  - Physical layer--agnostic
  - Data agnostic
- Syntactic
  - Data format/structure
  - Data translatability
- Semantic
  - Meaning of data – machine interpretation

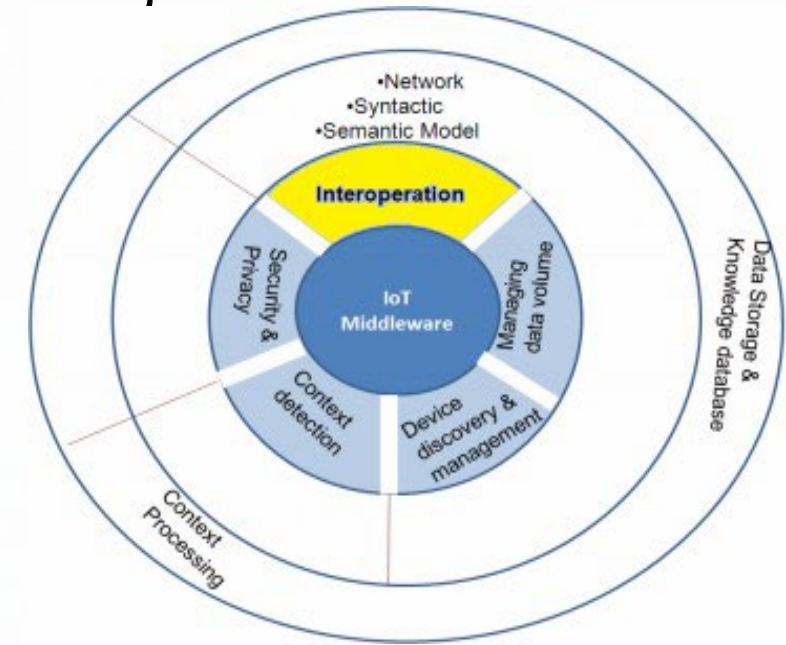


# Interoperation & Layering

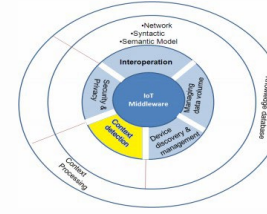
- APIs & abstraction: longstanding software engineering traditions



- Help make interoperability possible:
  - Separates domain responsibilities
  - Identifies requirements for modular implementation (API)
  - Builds upon established requirements/specifications



# Context Detection

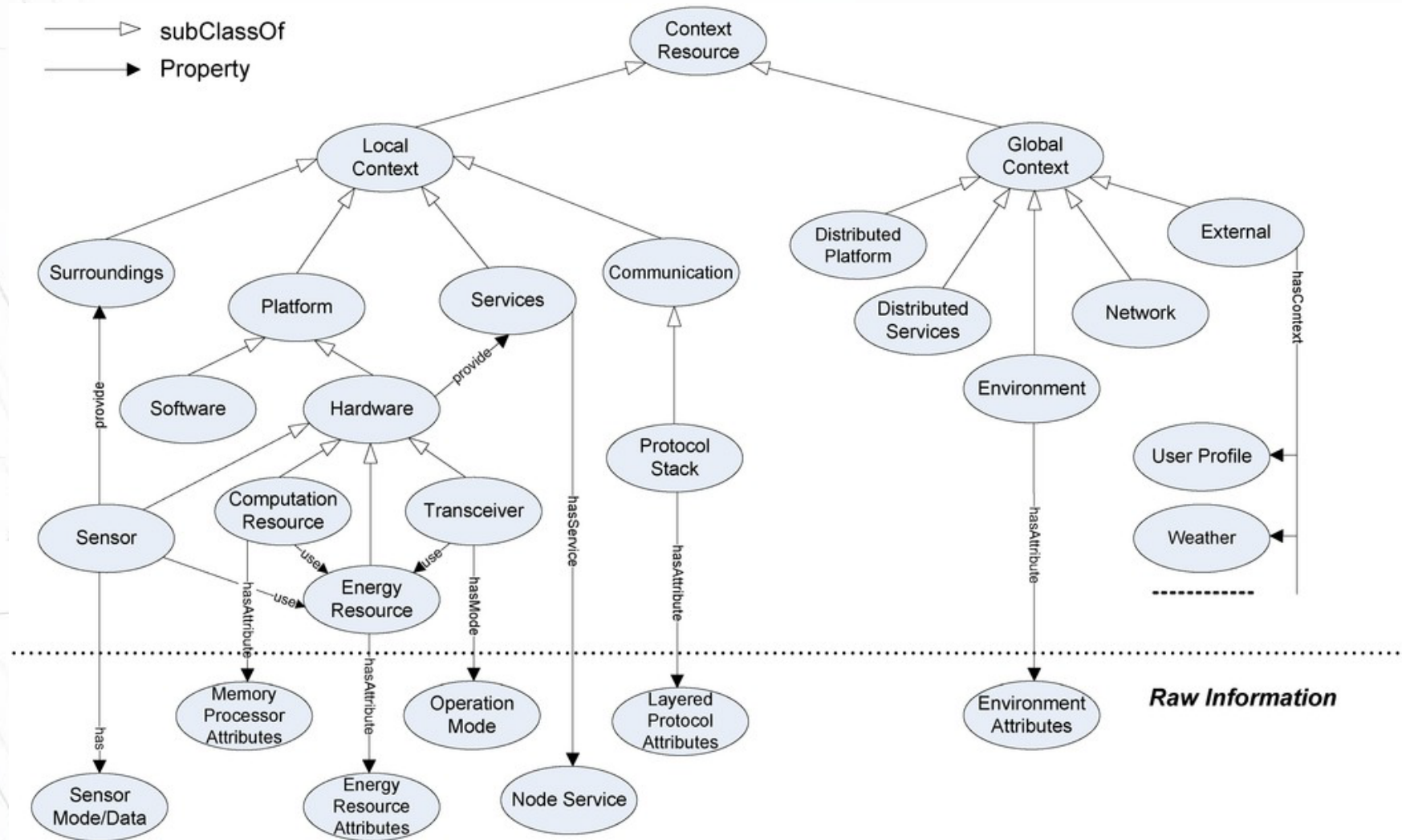


- Data → Knowledge
- Answer fundamental questions about data:
  - **What** does it mean?
  - **Who** needs to know about it?
  - **Why** is it important?
  - **When** did this happen?
- Metadata encapsulates data (more layering!)
  - Enables *automated consumption* by IoT applications





# Semantic Interoperability & Context

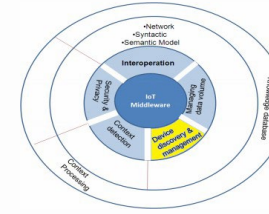




# Algorithms for Context Detection

- Typically fall under machine learning:
  - **Model generation:** formulate output based on a functional combination of input
  - **Discrete--event detection:** pattern matching, anomaly detection, clustering
  - **Supervised/Unsupervised learning** of output states
  - **State models:** decision trees, Bayesian networks

# Device Discovery



- Identify other nearby devices, and make yourself known
  - Semantic state – *device ontology*
- Virtual device interface
  - Identification/Association
  - Capabilities/RPCs
  - State information about rest of infrastructure
- Enables further communication
  - Transfer of information
  - Routing (P2P)
  - Application--specific implementation changes



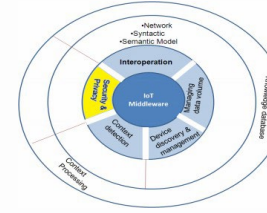
# Device Discovery Implementations

- Bluetooth Beacons
  - One--directional location/metadata detection
- WiFi Aware
  - One--directional identifier broadcast
  - Metadata can allow for ad--hoc, two--way communication
- Physical Web
  - One--way signal + URL for additional connectivity
- Protocols:
  - DNS--SD (service discovery)
  - Multicast DNS
  - W3C Network service discovery



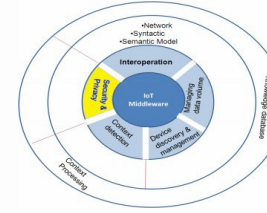


# Security & Privacy



- Fundamental problem:
  - Nodes need to discover other nodes, but still maintain privacy
- Middleware solution: multi-tier privacy (layering yet again!)
  - Management of discoverability, data access
  - Use device semantics to identify accessibility...but how to limit discoverability?

# Security Implementations



- Public Key Infrastructure
  - Requires backend access
- Group Key: Authorization before Inclusion
  - Requires new device to register one time with backend
- Local Access Control:
  - Local DNS
- Multi--stage access:
  - Public – advertise ID only
  - Semi--Private: Authorization allows metadata access
  - Private: Full access allows data access

# Scalability

## ■ Expected:

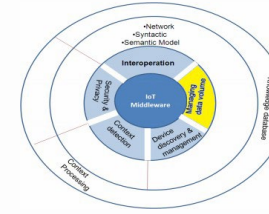
- Trillions of devices
- Exabytes of data

## ■ Current reality:

- Millions of devices
- Approaching petabytes of data

## ■ Fundamental scalability problems:

- **Storage:** persistent? raw? summarized?
- **Processing:** how much data to use?
  - how do applications scale with more data availability?
- **Communication:** how much data to transmit?
  - what infrastructure will scale to allow more data transmission while still meeting application deadlines?



# Middleware Implementation Resources

- <https://www.slideshare.net/slideshow/opensource-iot-middleware-frameworks/88268738>
- [https://cseweb.ucsd.edu/classes/wi16/cse291-c/Week7\\_Middleware\\_implementations.pdf](https://cseweb.ucsd.edu/classes/wi16/cse291-c/Week7_Middleware_implementations.pdf)
- Networking + Application
- [https://profile.iiita.ac.in/bibhas.ghoshal/IoT\\_2021/Slides/Application\\_Protocols.pdf](https://profile.iiita.ac.in/bibhas.ghoshal/IoT_2021/Slides/Application_Protocols.pdf)



# References

- <https://qiriro.com/ecs6264/lectures/>
- <https://cseweb.ucsd.edu/classes/wi16/cse291-c/>



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# Thank You

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